Mohamed Irshad Nadha

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Statement

I'm a final-year undergraduate with a strong interest in human-centered computing, focusing on machine learning and bio-signal processing. I am passionate about developing technologies that enhance human well-being, particularly in the healthcare domain. My goal is to design and build intelligent, user-centered solutions that can make a meaningful impact on people's lives.

Education

University of Moratuwa, Sri Lanka *Feb 2023 - Present*
B.Sc (Hons) in Electronic and Telecommunication Engineering
CGPA: 3.73/4.0
Minor in Mathematics: Graph Theory, Linear Algebra, Applied Statistics, Numerical Methods, Neural networks and fuzzy logic
Expected Graduation: 2027

Amina Girls' National School, Matale, Sri Lanka *Graduated: Jan 2022*
GCE Advanced Level (Combined Mathematics, Physics, Chemistry)
Achievements: 3A (distinction) passes, Ranked 334th in the country, Z-score: 2.3333
(Country-wide university entrance exam taken by over 100,000 students annually)

Certificates (External)

- Machine Learning Specialization – DeepLearning.AI and Stanford Online
- Mathematics for Machine Learning – DeepLearning.AI and Stanford Online
- Artificial Intelligence in Embedded Systems (Embedded ML for edge computing) – SkillSurf.lk and ENTC
- PyTorch Bootcamp – OpenCV University

Technical Skills

Programming Languages: Python, C, C++, MATLAB
Frameworks: PyTorch, TensorFlow, Keras
Other Skills: Altium Designer, SolidWorks, Multisim Simulator
Interests: Human-centered computing, machine learning and AI, bio signal processing

Soft Skills

Language Skills:

- **English:** Bilingual proficiency (speaking and writing)
- **Tamil:** Native proficiency (speaking and writing)
- **Sinhala:** Limited working proficiency (speaking and writing)

Other Skills: Strong writing, communication, interpersonal, leadership, analytical, and problem-solving skills

Research Experience

- **Visiting Research Student**
  School of Computer Science, The University of Sydney Dec 2025 - June 2026
  Supervisor: Associate Prof. Anusha Vithana
  Research area: Human-Computer Interaction
  - **Project 1: Australian Sign Language (Auslan):** Contributed to data collection and evaluation for a project developing an Australian Sign Language (Auslan) dataset to help hearing parents learn to communicate with their deaf children.
  - **Project 2: Human Motion Prediction:** Worked on predicting human motion using IMU sensors placed at different locations on the body, applying signal processing and machine learning techniques to analyze motion patterns.
- **Research Contributor**, LiverUSRecon Research Team July 2025 - Nov 2025
  Supervisor: Prof. Ranga Rodrigo
  Project: Automatic 3D Reconstruction and Volumetry of the Liver with a Few Partial Ultrasound Scans
  - Contributed to medical image analysis research on 3D organ reconstruction and volumetry.
  - Tested multiple datasets for **liver ultrasound reconstruction** and conducted experiments to extend the approach to the **kidney**.
  - Initial experiments conducted with the **TransUNet** model followed by **LeViT-UNet** for improved performance.
- **Research Intern**, Singapore University of Technology and Design (SUTD) July - Nov 2025
  Supervisor: Prof. Na Zhao
  Topic: 3D Domain Generalization for Autonomous Driving
  - Conducted research on domain generalization for 3D object detection in autonomous driving.
  - Worked with the **nuScenes** and **Argoverse 2** datasets using the **CMT** and **FSDv2** codebases, focusing on cross-domain robustness.

Projects

- **Australian Sign Language (Auslan) Dataset and Recognition** (Dec 2025 - June 2026)
  - Led the end-to-end development of an Australian Sign Language (Auslan) dataset, overseeing data collection, coordination with external parties, and organization of the overall data collection process.
  - Developed and evaluated deep learning models for sign language recognition, comparing multiple architectures using the collected dataset and analyzing their performance.
  - Conducted experiments to evaluate model performance and dataset characteristics, co-authoring a conference paper currently under peer review.
- **MotionMAE – Human Motion Prediction** (March - June 2026)
  - Studied and evaluated existing human motion prediction models using IMU data collected from sensors placed at different body locations.
  - Conducted sensitivity and noise analyses to investigate the impact of sensor placement and measurement noise on human motion prediction performance.
  - Performed ablation studies to evaluate the contribution of different components and configurations of the motion prediction framework.
- **Bin Picking Robot** (Jan - July 2025)

  - Designed and implemented the vision system for a Cartesian robot with a suction gripper to perform object picking from bins.
  - Evaluated multiple instance segmentation models; selected and deployed YOLOv11 for its high accuracy and efficiency on edge hardware.
  - Handled camera placement optimization and calibration to align vision output with the robot's coordinate frame for precise object localization.
- **GlucoPal - Non-Invasive Glucometer using NIR and PPG Signals** (March - July 2025)
  - GlucoPal is a non-invasive glucose monitoring system using near-infrared (NIR) sensors and photoplethysmography (PPG) signals.
  - Captured raw PPG signals, applied biosignal processing techniques (filtering, normalization, feature extraction) to enhance signal quality and extract glucose-relevant patterns.
  - Designed and trained a ResNet34-based deep learning model in PyTorch to predict blood glucose levels from processed PPG waveforms.
- **FGVC12 Challenges @ CVPR 2025**
  - Participated in a series of cutting-edge computer vision competitions focused on fine-grained visual categorization (FGVC), hosted as part of the CVPR 2025 FGVC12 Workshop.
  - Applied deep learning techniques to tasks such as species identification, plant recognition from quadrat images, and animal re-identification.
  - Explored datasets like AnimalCLEF25, FungiCLEF25, and PlantClef 2025, advancing research in biodiversity and ecological monitoring.

Volunteer Experience & Leadership

- **Manager, External Relations**(25/26) - Electronic Club, UoM
- **Assistant Secretary** (24/25) - Mathematics Society, UoM
- **Director of Membership Development** (23/24) - Mathematics Society, UoM
- **Student Representative** - Department of Electronics and Telecommunications (Semesters 2 and 3)

Referees

Prof. Ranga Rodrigo
B.Sc. Eng.Hons (Moratuwa), M.E.Sc. (Western, Canada), Ph.D. (Western, Canada)
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